

Observer-Independent Method for Microstructural Parcellation of Cerebral Cortex: A Quantitative Approach to Cytoarchitectonics¹

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We describe a new, observer-independent procedure for identifying boundaries between cortical areas. The method is useful for images obtained from sections which provide microstructural information on the cortical laminar pattern, e.g., Nissl-, myelin-, or immunohistochemically stained sections or receptor autoradiographs. The laminar pattern is represented by profile curves extending from the cortical surface to the white matter boundary. These profiles are constructed from digitized images. Digitization is based on the grey level index (Nissl) or densitometry (myelin, immunohistochemistry, receptor autoradiography). The shapes of neighboring profiles are compared by calculating their distances according to feature vectors extracted from the profiles. Profiles derived from a homogeneous area can be expected to be similar in shape and hence show low distance values between each other. Maximum distances can be found between profiles which lie on opposite sides of a structural boundary. The Mahalanobis distance was found to be more sensitive and to yield greater spatial resolution than other distance measures such as the Euclidean distance. Cell-stained sections of the human neocortex were analyzed. The method not only verified boundaries which had been defined by visual inspection, it also revealed new ones which had not been detected visually. The procedure offers an important supplement to the traditional methods based on visual inspection which, for the first time, is based on quantitative data and therefore offers a new level of reproducibility and observer independence. Anatomical atlases based on this procedure thus provide a new tool for the interpretation of structural data obtained from functional imaging techniques. © 1999 Academic Press

Key Words: brain mapping; cerebral cortex; cytoarchitecture; grey level index; density profiles; multivariate statistics.

INTRODUCTION

The microstructural parcellation of the cerebral cortex has been a challenge for generations of neurobiologists (for references see Schleicher and Zilles, 1989). Most earlier studies were based on the visual, microscopical inspection of histological sections. The qualitative nature of this approach has contributed to differences between the maps of different authors in terms of number, localization, and contours of cortical areas (e.g., Brodmann, 1909; von Economo and Koskinas, 1925; Lashley and Clarke, 1946; Bailey and v. Bonin, 1951).

Such marked differences can be explained in part by various factors not related to the mapping procedure. Different brains were used to establish the maps, and the interindividual variability of the areal pattern in the neocortex is remarkably high (Rademacher *et al.*, 1993). An additional source of variability may be introduced by differences in display modes and viewing angles. The amount of variability attributed to the mapping procedure itself is unknown and might only be determined by mapping the same brain by different investigators independently using their own parcellation criteria. There is ample evidence that not only the position of distinct boundaries is subject to observer-dependent criteria, but also the degree of parcellation is not consistent among investigators. For example, areas 44, 45, and 47 in the map of Brodmann (1909) correspond only in part to 3 areas by von Economo and Koskinas and these are supplemented by a series of transitional areas in the fronto-orbital cortex, as, e.g., F_{FF}, F_F, and F_{fa} (von Economo and Koskinas, 1925). The same 3 areas correspond to 10 areas (57–66) by Riegele (1935), whereby his areas cannot definitely be associated with corresponding areas in the other 2 maps. A more recent cytoarchitectonic study which was based on the nomenclature of Brodmann added a new transitional area 45–9 (Rajkowska and Goldman-Rakic, 1995). Thus, these authors not only used different criteria for establishing the same cytoarchitectonic boundary, they applied a completely different classifica-

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tion scheme with different hierarchical principles. Discrepancies of this type can only be solved by using formalized, i.e., quantified criteria for defining cytoarchitectonic parcellation patterns.

The lack of a standard methodology with which to establish cortical boundaries on the basis of detectable differences in the structural composition of areas can be explained by the complex texture of the cellular arrangement. Since the cortex is organized vertically into columns, modules, and dendritic bundles, as well as horizontally in layers parallel to the pial surface, a variety of parameters including cell types and shapes, cell density, and pattern of arrangement of cells are suitable to describe the structural inhomogeneity within a cortical area and differences between areas. Some of these inhomogeneities cause a laminar pattern to become visible in various preparations such as Nissl- or myelin-stained sections and in autoradiographs. It is a widely accepted finding that the functional differentiation of the cortex into areas is reflected by a specific structural organization and thus by a specific laminar pattern. Therefore, a straightforward strategy with which to establish a boundary between areas is to define it as the point at which the laminar pattern changes. A classical example of a boundary is the point at which there is an abrupt decrease in the thickness of layer IV at the border between the primary visual cortex V1 (Brodmann's area 17) and the adjacent area V2 (Brodmann's area 18). However, only a few boundaries are as obvious to visual inspection as this one. Most other boundary definitions depend on the set of specific parameters used, on its qualitative evaluation and the experience of the investigator. It was expected of the computerized image analyzing techniques that they resolve this lack of consistency.

Cortical maps of numerous species were drawn or revised as computerized image analyzing systems were developed. These systems facilitated the analysis of structural inhomogeneities in large samples (Fleischhauer *et al.*, 1980; Rehkämper *et al.*, 1984; Wree *et al.*, 1981, 1983; Zilles *et al.*, 1978, 1980, 1982). The techniques were based on substantially improved imaging of the laminar patterns rather than on purely qualitative, visual inspection. The most crucial step, however, namely the identification of cortical boundaries was still left to the interpretation of the investigator and could not be automated. Due to this unsolved problem, observer-dependent variability as to the definition of cortical areas remained. Since earlier procedures localized the boundary of a brain region by analyzing changes in cell density, they could not be applied to the detection of boundaries in the cortex, which is, due to its laminar organization, inhomogeneous in cell density. Automated procedures for the delineation of brain regions were thus limited to subcortical structures (for references see Hoogendijk *et al.*, 1995).

More advanced automated procedures have become established for quantifying the laminar pattern by density profiles representing the course of a density measure (e.g., volume density of neurons, concentration of myelin, receptor density, density of immunopositive structures). These profiles extend from the pial surface to the white matter boundary. Profiles extracted from different cortical areas have different shapes and this feature has been used to characterize cortical layers (Amunts *et al.*, 1995) or to compare cortical areas (Geyer *et al.*, 1997; Hudspeth *et al.*, 1976; Sauer, 1983a,b). Profiles have also been used to analyze left-right differences between the hemispheres of the same brain (Amunts *et al.*, 1996, 1997b).

Anatomical brain mapping has gained in importance with the advent of computerized brain atlases. A new type of human brain atlas has recently been established which includes data from not only one brain, but constructs probabilistic maps compiled from several brains. The atlas can therefore take the high degree of interindividual variability between human brains into account (Mazziotta *et al.*, 1995). When trying to integrate areas defined on a microstructural basis into a probabilistic atlas so that data obtained from functional imaging studies can be interpreted, one problem becomes obvious: cortical regions are highly variable in size, shape, and position. This variability among individuals is due to two major factors: true interindividual variability and variability introduced by the subjective definition of cortical boundaries on a microstructural level. In order to study interindividual variability in structure, it is necessary to eliminate variability introduced by the observer.

Our study presents for the first time an automated and observer-independent procedure for parcellating the cerebral cortex which combines shape description of density profiles with multivariate distance analysis.

MATERIAL AND METHODS

Histology and Image Acquisition

The procedure for locating boundaries was tested for areas V1, V2, and VP in the human occipital cortex and for areas Te1.0, Te1.1, Te1.2, and Te3.0 in the temporal cortex. Coronal, 20- μ m-thick sections of formalin-fixed, paraffin-embedded tissue were processed with a silver staining technique (Merker, 1983). Data were acquired from rectangular regions of interest (ROI) at low microscopic resolution (Planapo 6.3 $\times$ 1.25, Zeiss, Germany; Fig. 1). For details, see Schleicher and Zilles (1989). In short, using an automatic scanning procedure, the areal fraction of darkly stained cell bodies is measured in square, adjoining measuring fields (e.g., 25 μ m per side) after thresholding. This areal fraction is an esti-

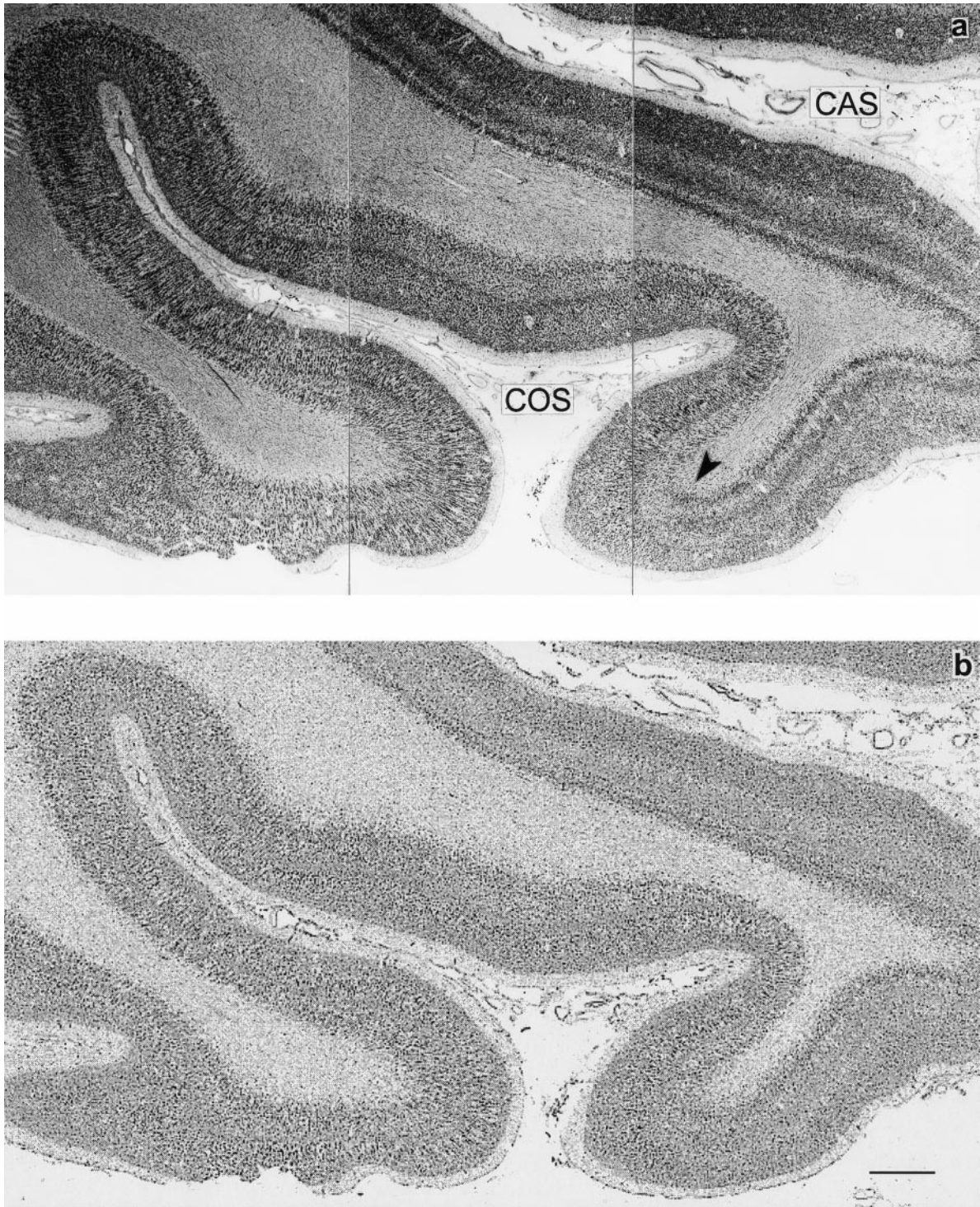


FIG. 1. (a) Photomicrograph of silver-stained, coronal section of adult human brain showing the primary visual cortex V1 and the adjoining areas (CAS, calcarine sulcus; COS, collateral sulcus). Arrowhead marks boundary between V1 (right) and V2 (left). (b) GLI image of (a), scanned at a spatial resolution of 25 μm . The GLI values are scaled to the range from white (GLI = 0%) to black (GLI = 50%) for printing. Bar, 1 mm in both.

mate of the volume density of cell bodies and will henceforth be referred to as "GLI" (Grey level index; $0\% \leq \text{GLI} \leq 100\%$). The GLI is highly correlated with the volume density of neurons (Wree *et al.*, 1982), since the presence of endothelial and of glial cells is not dependent on the laminar location in the cortex and may therefore be interpreted as a constant, additive contribution to the volume density of neurons.

The ROI is completely scanned in a mosaic-like pattern, and the GLI is determined for each field. The resulting data matrix is the GLI image. GLI scans are superior to TV scans, even to those acquired with high resolution cameras, since variations in staining intensity of cell bodies or background will not influence the GLI values if the cell bodies can be segmented from the background.

Density Profiles

The procedure for locating boundaries is based on the finding that each area has its own typical laminar pattern which is different from that of neighboring areas. Such differences are the prerequisite for defining a cytoarchitectonic or myeloarchitectonic area. We characterized the laminar pattern on the basis of density profiles extracted along traverses extending from the border between layers I and II to the border between layer VI and the white matter (Fig. 2). In order to align the traverses orthogonally to the laminae, a line was drawn midway between layers I/II and the layer VI/white matter border, and the traverses transected this line perpendicularly. The traverses were placed interactively by entering the start and end points using a

graphics tablet. A fully automated software program for positioning traverses and extracting profiles is under development.

Using the traverses, the ROI was covered by a set of profiles which were equally spaced at intervals of approximately 200 μm (Fig. 4a). A spacing interval of 200 μm reduced spatial resolution to cytoarchitectonically defined areas and excluded smaller subunits such as cortical columns (Mountcastle, 1978) from detection. In addition, each GLI image was smoothed by two passes of a 3×3 average filter prior to extracting the profiles in order to reduce the effect of varying cell densities caused by cortical columns. Columns are about 40 μm wide and are oriented parallel to the traverses. Unfiltered images caused periodic variations in GLI values (Schlaug *et al.*, 1995) and consequently increased the differences in shape between neighboring profiles. Each profile was standardized to encompass 101 values corresponding to a cortical depth from 0 to 100% by resampling the data using linear interpolation (Fig. 3a). The routines for generating GLI images and for extracting and processing profiles were carried out with a KS400 image analyzing system (Zeiss, Germany).

Feature Vector and Procedure for Locating Boundaries

Maximum differences in profile shape can be expected at locations at which each of two neighboring profiles was extracted from a different area, i.e., two profiles face a boundary between two areas.

Profile shape was quantified using a set of 10 features f_i ($1 \leq i \leq p$, p = number of features) extracted

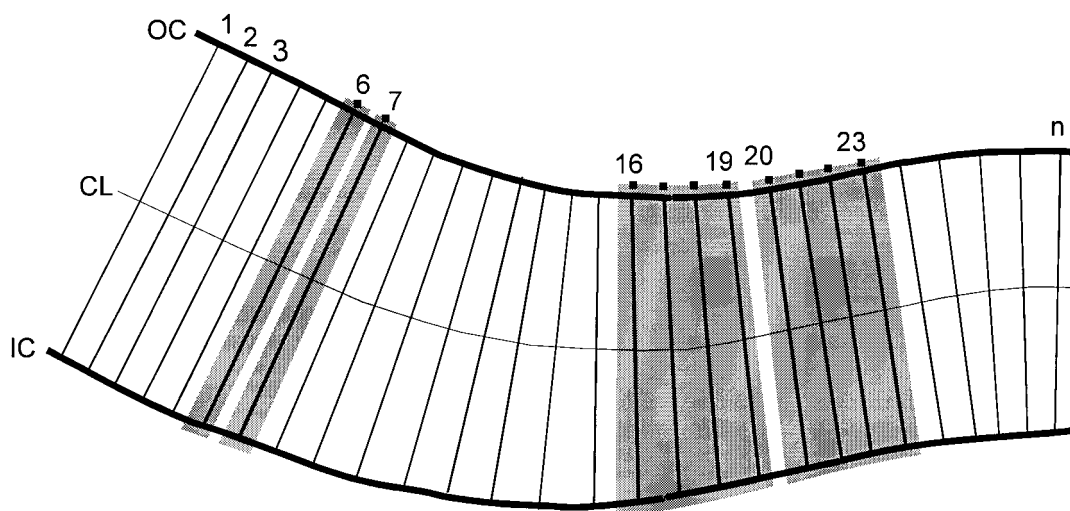


FIG. 2. Positioning and grouping of density profiles. OC, outer contour, i.e., pial surface or border between layers I and II; IC, inner contour, i.e., border between cortex and white matter; CL, central line. 1 to n , positions of profiles ($1 \leq j \leq n$). Distances are calculated between two neighboring profiles ($b = 1$, e.g., $j = 6$ and $j + 1 = 7$, shaded areas, distance value is assigned to position $j = 6$) or between two neighboring blocks of profiles (blocks of $b = 4$ profiles in this diagram: block 1 has profiles 16–19, block 2 has profiles 20–23; shaded areas, distance value is assigned to position $j = 19$). At this position, the mean feature vector X_j (profiles 16–19), the mean feature vector X_{j+1} (profiles 20–23) and the pooled covariance matrix $C_{j,j+1}^{-1}$ from the profiles 16 to 23 were used for the calculation of the Mahalanobis distance.

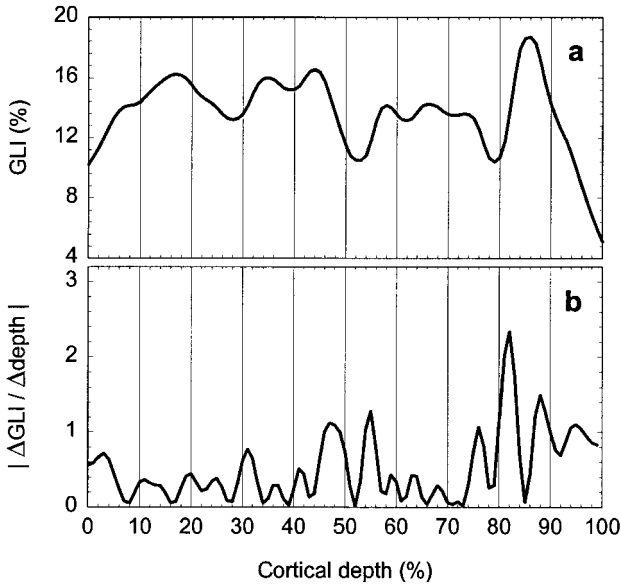


FIG. 3. (a) Density profile describing the course of the GLI (ordinate) from the border between layers I and II (cortical depth = 0%) to the border between layer VI and white matter (cortical depth = 100%). The profile is located in the primary visual cortex at position $j = 135$ in Fig. 4a. (b) Absolute value of first derivative (differential quotient) of the profile shown in (a).

from each profile and from the absolute value of its first derivative (differential quotient). The features were the mean amplitude and, by interpreting both signals as frequency curves, the first four central moments m_k about the mean ($1 \leq k \leq 4$) (Dixon *et al.*, 1988). m_1 is the mean, m_2 is scaled to the standard deviation, m_3 to the skewness, and m_4 to the kurtosis (Table 1). The five features of each profile and the five features from its differential quotient were combined into one feature vector $\mathbf{X}$. The data from the sample of n profiles of an ROI were standardized to z scores in order to assign equal weight to each of the features.

Differences in shape between two profiles were calcu-

TABLE 1

Feature Vector of the Profile Shown in Fig. 3

	Original data		z scores	
	Profile	Derivative	Profile	Derivative
Mean amplitude	13.699	0.516	-0.984	1.714
m_1	48.410	60.404	-0.648	2.877
m_2	28.269	29.723	1.311	-1.590
m_3	0.056	-0.529	0.351	-2.531
m_4	-1.209	-0.962	-1.165	-2.753

Note. Two data sets are shown: original data and z scores calculated by standardizing the feature values within the set of 135 profiles of Fig. 4. Features of the profile and of the analogous features of absolute value of its first derivative are combined into one feature vector.

lated as the distance between two p -dimensional feature vectors. The Euclidean distance between the feature vectors $\mathbf{X}_j$ and $\mathbf{X}_{j+1}$ ($j = \text{position of profile}; 1 \leq j \leq (n - 1)$) is defined as

$$E_j^2 = (\mathbf{X}_j - \mathbf{X}_{j+1})' \cdot (\mathbf{X}_j - \mathbf{X}_{j+1}). \quad (1)$$

The distance function $DE^2(j)$ was established by calculating the distances between all pairs of neighboring profiles and plotting the values as a function of j (Fig. 4b).

Although the GLI images were smoothed, profiles were affected by local structural inhomogeneities such as cell clusters or blood vessels. The signal-to-noise ratio of the distance function therefore had to be improved by calculating the distance between two adjacent blocks of profiles instead of the distance between two single adjacent profiles. Each block comprises a b number of neighboring profiles. $\mathbf{X}_j$ denotes the mean feature vector of the block at position j , $\mathbf{X}_{j+1}$ the mean feature vector of the adjacent block at position $j + 1$ (Fig. 2, for result see Fig. 4c). For $b \geq p/2 + 1$, the Mahalanobis distance

$$M_j^2 = (\mathbf{X}_j - \mathbf{X}_{j+1})' \cdot \mathbf{C}_{j,j+1}^{-1} \cdot (\mathbf{X}_j - \mathbf{X}_{j+1}) \quad (2)$$

(Mahalanobis *et al.*, 1949) was calculated as an additional measure of distance. $\mathbf{C}_{j,j+1}^{-1}$ ($b \leq j \leq n - b$) denotes the inverse of the pooled covariance matrix of the feature values from both blocks at positions j and $j + 1$ (Fig. 4d).

Distance functions showed numerous maxima of various heights. In the case of the Mahalanobis distance, the Hotelling's statistic

$$T_j^2 = M_j^2 / (1/n_1 + 1/n_2) \quad (3)$$

(Dixon *et al.*, 1988) with $n_1 = n_2 = b$ was calculated to classify the maxima. Since the profiles within a ROI were not independent and no correction for multiple comparisons was performed, the corresponding P value was not used in a statistical test but as a variable to group maxima into categories of differing relevance.

The distance function should yield maximum values at positions marking the interface between two adjacent blocks of profiles, each of which belongs to a different area and contains profiles from only that area. Thus, the spatial resolution of this procedure is limited by the size of the blocks. Only areas as wide or wider than the cortical sector covered by one block can be resolved with correct distance values. Maxima of the Mahalanobis-based distance function $DM^2(j)$ were therefore processed according to the following conventions: (1) Discard maxima with $P > 0.05$. (2) Compare all pairs of neighboring maxima. If their differences in

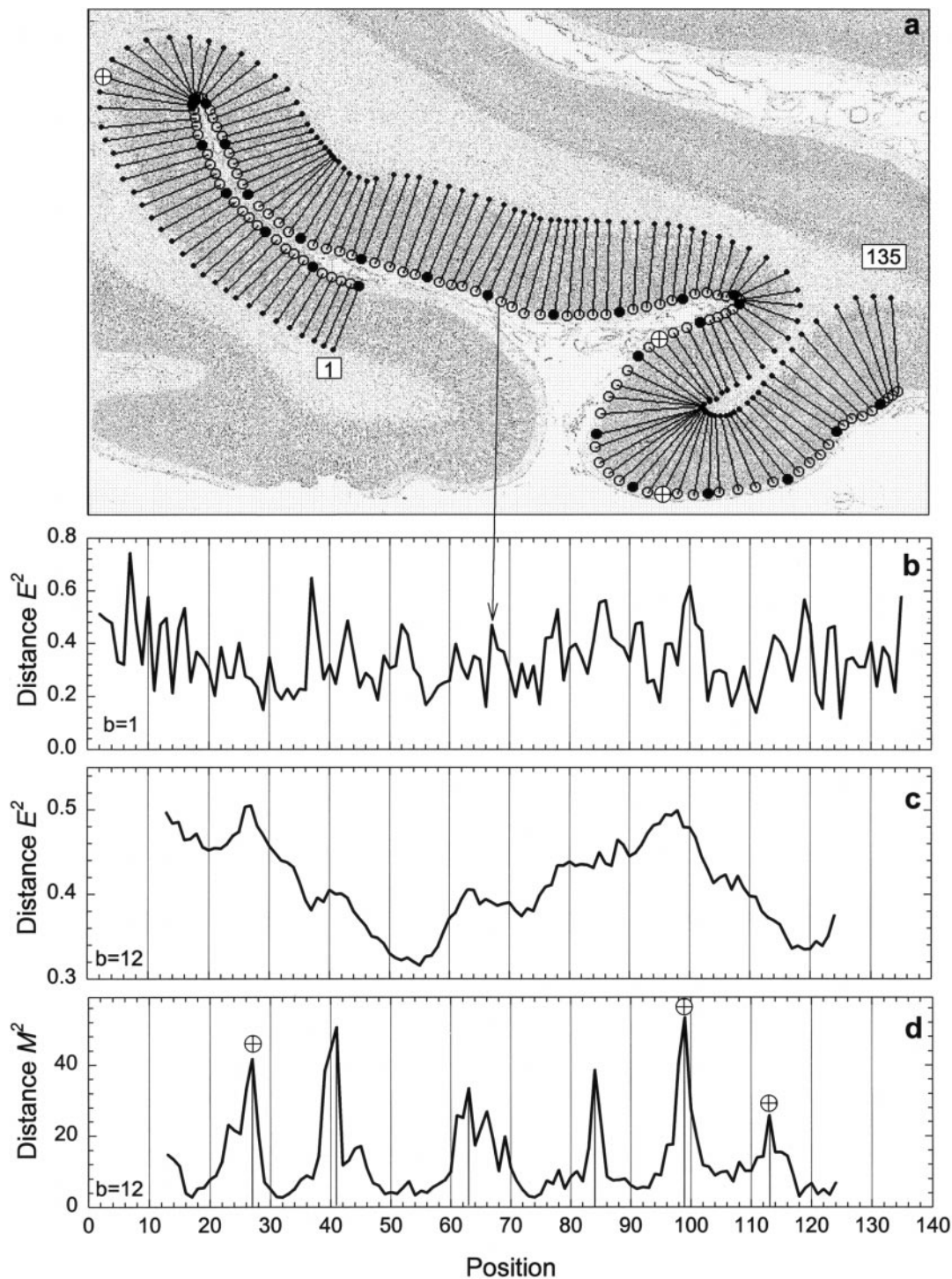


FIG. 4. (a) GLI scan of the region shown in Fig. 1, reduced in contrast, with a set of 135 superimposed traverses which cover areas V1, V2, and VP. Starting points of traverses are marked by circles, endpoints are marked by dots. Numbering j of profiles ($1 \leq j \leq 135$) begins on the left side of the GLI scan, positions 1, 6, 11, 16 . . . are marked by filled circles. Crossed circles mark positions 27 (border between VP and V2), 99 (border between V2 and $V2_{OBY}$), and 113 (border between $V2_{OBY}$ and V1). (b) Euclidean distance function $DE^2(j)$ for set of profiles shown in (a) calculated for each pair of neighboring individual profiles ($b = 1$). The arrow shows an example: it starts at profile $j = 67$ in (a) and points to the corresponding distance value d_{67} between profiles 67 and 68 shown in (b). (c) Euclidean distance function $DE^2(j)$ calculated for each pair of neighboring blocks of profiles ($b = 12$ profiles per block). (d) Mahalanobis distance function $DM^2(j)$ for each pair of neighboring blocks of profiles ($b = 12$ profiles per block). Main maxima are denoted by drop lines. Crossed circles label the maxima with corresponding positions marked in (a).

position are smaller than the size of one block, discard the maximum with the higher P value. The procedure converged after a second pass of step (2). The remaining maxima are called main maxima and were spatially separated by at least the size of one block (Fig. 4d).

Simulation of Cortical Layering

In order to compare Euclidean and Mahalanobis distance functions with respect to spatial resolution and sensitivity for detecting predefined boundaries, a model of a layered cortical region was generated (Fig. 5). Five layers were represented by horizontal stripes with differing GLI values (I, 12; II, 63; III, 12; IV, 39/25; V, 12) and random noise was added (normally distributed, signal-to-noise ratio $\text{SNR}_{\text{rms}} = 8.2$; Gonzales and Wintz, 1987). Between positions 119 and 181, a sine-shaped thinning of layer II was inserted to represent a distortion in the thickness of layers II and III induced by cortical folding. Boundaries were predefined in layer IV at positions 30, 60, 68, 100, and 168 by step functions in GLI values. The distance between positions 60 and 68 was used to explore the dependency of spatial resolution on b , the number of profiles per block. Euclidean distance functions $DE^2(j)$ and Mahalanobis distance functions $DM^2(j)$ were generated for $b = 8$ and $b = 12$.

RESULTS

Mapping of Visual Areas

The boundary between visual areas V1 and V2 is a classical example of a change in laminar patterns at the interface between two cortical areas. When viewing the histological section (Fig. 1), this boundary could easily be found. The Euclidean distance function based on distances between two neighboring profiles ($DE^2(j)$, $b = 1$; Figs. 4b), however, did not selectively highlight this nor any other boundary due to the large number of maxima which could not be assigned to distinct changes in the laminar pattern. For $b = 1$, similar results were found in other sections and tissue blocks irrespective of cortical region. Distance functions for neighboring blocks with $b = 12$ profiles were therefore calculated. Surprisingly enough, ($DE^2(j)$) did not indicate this boundary either (Fig. 4c). The Mahalanobis distance function $DM^2(j)$, however, clearly indicated the boundary at position 113 (Fig. 4d).

The location of the boundary between V2 and VP is much less obvious visually than the border between V1 and V2. Two main maxima in $DM^2(j)$, positions 27 and 41 (Fig. 4d), clearly marked microstructural boundaries and were possible candidates. Visual inspection showed that position 27 possessed the cytoarchitectonic characteristics as described by Clark (1993). Both main maxima could also be found in $DE^2(j)$ (Fig. 4c).

Additional main maxima in $DM^2(j)$ were located at positions 63, 84, and 99. Position 99 was found to delineate subarea $V2_{\text{OBy}}$ from the rest of V2. We did not try to assign maxima at positions 41, 63, and 84 to functionally defined subareas within V2, but there is strong evidence that a subparcellation of V2, as demonstrated in cytochrome oxidase preparations, corresponds to a functional subdivision of V2 (Wong-Riley et al., 1993).

Mapping of the Simulated Model

The characteristics of $DE^2(j)$ and $DM^2(j)$ were further examined by analyzing the model (Fig. 5). For $b = 8$ (Figs. 5b and 5c), both distance functions unambiguously detected all boundaries which had been predefined in layer IV. $DM^2(j)$ (Fig. 5c) yielded sharp maxima but did not detect the sine-shaped distortion between layers II and III, whereas $DE^2(j)$ (Fig. 5b) yielded broad peaks and clearly detected the distortion. The maximum at position 134 in Fig. 5b was caused only by the distortion. These differences in the sensitivity of the distance functions became even more obvious when the distance functions for $b = 12$ were compared (Figs. 5d and 5e). In $DE^2(j)$ (Fig. 5d), the maximum at position 134 could not be distinguished from the other maxima, but it was not identified as a main maximum in $DM^2(j)$ (Fig. 5e). The spatial resolution of the two distance functions differed. This is demonstrated by the stretch of cortex between positions 60 and 68: $DE^2(j)$ indicated these boundaries for both values of b . If b was chosen at greater than 12 (data not shown), the peaks in $DE^2(j)$ became broad and failed to show clearly defined maxima. $DM^2(j)$ showed two separate peaks at positions 60 and 68 for $b = 8$ (Fig. 5c), but these were lost at $b = 12$ (Fig. 5e).

Summarizing these findings, one can state that $DM^2(j)$ was superior to $DE^2(j)$. $DM^2(j)$ was more sensitive to cortical boundaries (see the one between V1 and V2, mapping of visual areas above). $DM^2(j)$ was not, or to a much lesser degree, affected by distortions (low frequency shifts) in layer width such as may be caused by cortical folding. It is also preferable from a statistical point of view, since the Mahalanobis distance accounts for the correlations between features by weighting them with the elements of the covariance matrix (see Eq. (2)).

The Significance of Cortical Boundaries

The number and positions of the main maxima of the distance function depend on the value of b , i.e., the number of profiles per block. Distance functions at $b = 1$ were noisy and failed to show maxima indicative of boundaries. In $DE^2(j)$, the minimum value for b is 1. In $DM^2(j)$, however, the minimum value for b is related to the number of features p by the constraint $b \geq p/2 + 1$.

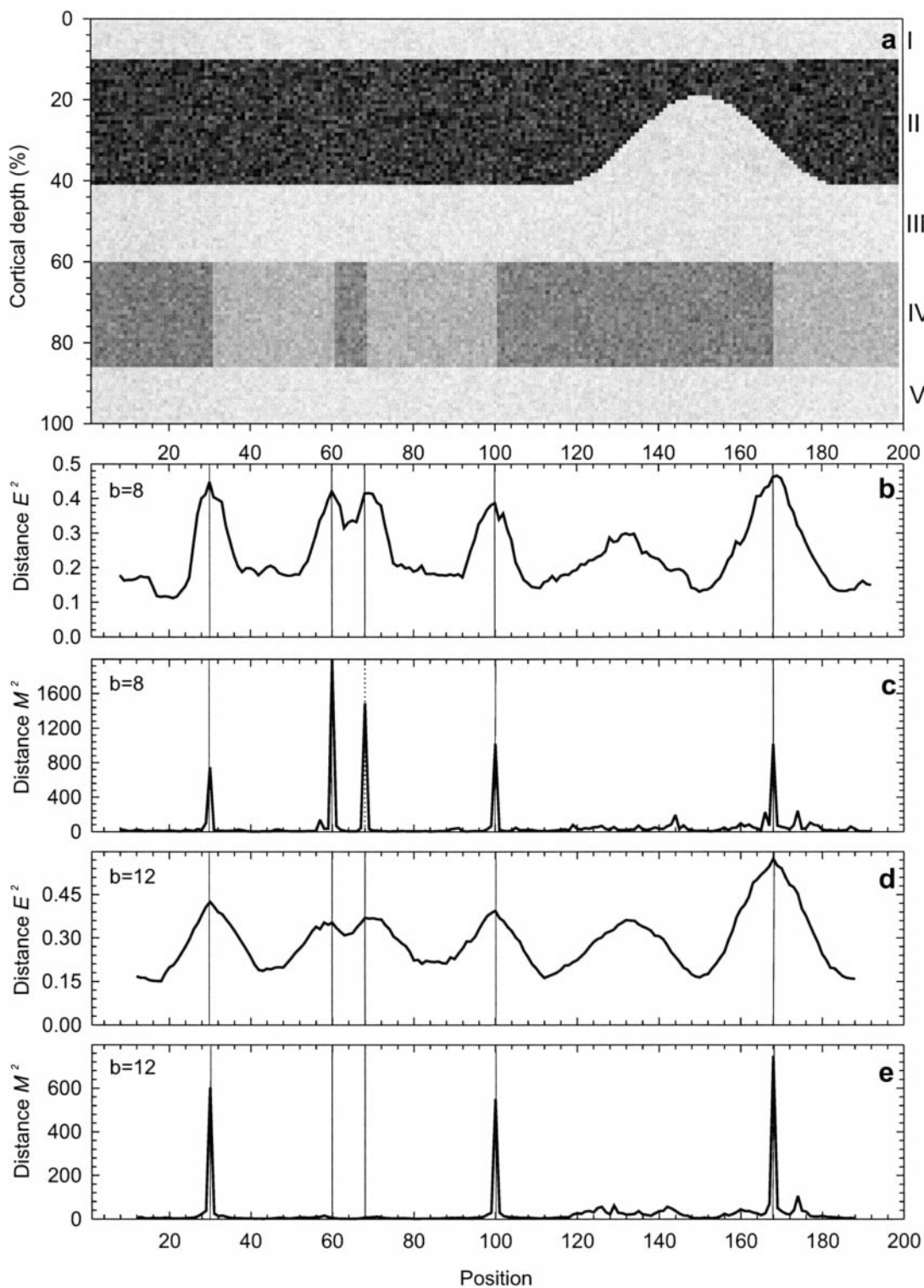


FIG. 5. (a) Model of a cortical region (200×100 pixel) with 5 layers (I–V). Sine-shaped distortion between layers II and III, cortical boundaries defined by changes in GLI values in layer IV. Each pixel column of the image was extracted as one profile. Abscissa in (a–e): position j of the 200 profiles. Ordinate in (a): cortical depth in %. (b) Euclidean distance function $DE^2(j)$, $b = 8$. Ordinate: distance E^2 . (c) Mahalanobis distance function $DM^2(j)$, $b = 8$. Ordinate: distance M^2 . (d) Euclidean distance function $DE^2(j)$, $b = 12$. Ordinate: distance E^2 . (e) Mahalanobis distance function $DM^2(j)$, $b = 12$. Ordinate: distance M^2 .

For our approach ($p = 10$), at least 6 profiles per block were required. Large values for b resulted in smooth distance functions, but they reduced the spatial resolution of the procedure, since a meaningful distance value was only achieved for each block of profiles lying completely within one area. We therefore explored to what extent the detection of main maxima depends on b . For the set of $n = 135$ profiles shown in Fig. 4a, b was increased from 6 to 18 (Fig. 6). For $b = 6$ and 7, the positions of the main maxima varied considerably, but only minor shifts occurred in the range of $b = 8$ to 15. For $b > 15$, increasingly fewer boundaries became detected because the spatial resolution of the algorithm was reduced. Interestingly enough, the positions of the main boundaries (between V2 and VP at position 27/28 and between V1 and V2 at position 113) were highly constant. Only the boundary between V2 and VP varied between positions 27 and 28. In our example, $b = 12$ seemed to be a good compromise between adequate resolution and smoothness of the distance function.

Since profiles acquired from one section cannot be regarded as statistically independent, Hotelling's T^2 to check for the significance of main maxima would artificially inflate the P values. Main maxima should therefore always be validated by comparing corresponding

maxima in a series of distance functions from adjoining sections. Since the probability of locating such maxima at comparable sites in a set of sections is exceedingly small, this procedure helps to exclude maxima which may be caused by artifacts (e.g., ruptures, folds) or local events like blood vessels or untypical cell clusters.

The mapping of the second transverse sulcus in the human temporal cortex using sagittal, silver-stained sections is shown as a second application (Fig. 7). We chose this region since its boundaries are difficult to be recognized visually. GLI measurements, profile extractions, and analyses were similar to those used for the visual areas. In order to be able to compare the positions of main maxima across sections despite changes in the shape of the cortical surface and variations in the spacing increment between traverses, a flat map was constructed by transferring each profile position to the distance of its traverse from the bottom of the second transverse temporal sulcus (Fig. 7). This distance was established by measuring the length of a line, which began at the bottom of the sulcus, followed the starting points of the traverses, and ended at the starting point of the traverse of the profile. For an example see the legend to Fig. 7.

Corresponding boundaries (detected by $DM^2(j)$, $b = 10$) were found at comparable positions across the three sections. For smaller values of b (data not shown), some additional boundaries were detected, but these were not consistent across the series and could therefore easily be recognized as random events.

DISCUSSION

Cutting effects. The angle of sectioning relative to orientation perpendicular to the pial surface affects the GLI profiles and thus the performance of our procedure. Two aspects can be considered: (i) *Analyzing an existing series of sections.* Since the laminar pattern, the cellular composition, and the local curvature vary greatly among cortical areas, no general rule can be given to determine the largest tolerable deviation from vertical sectioning. According to our experience, as much as a twofold increase in cortical thickness due to the oblique cutting does not distort the overall shape of density profiles. This corresponds to a plane of sectioning deviating 60° from the vertical. The absolute cortical depth is compensated for by scaling the profile length to 100% prior to extracting the shape-describing parameters. (ii) *Optimizing the sampling scheme.* Most stereological procedures designed for analyzing anisotropic structures require an optimized orientation of the plane of sectioning, which can only be established by a specific sampling scheme. It must clearly be stated that our procedure depends on a plane of sectioning which completely intersects all layers. Tangentially cut cortical regions with missing layers must be excluded.

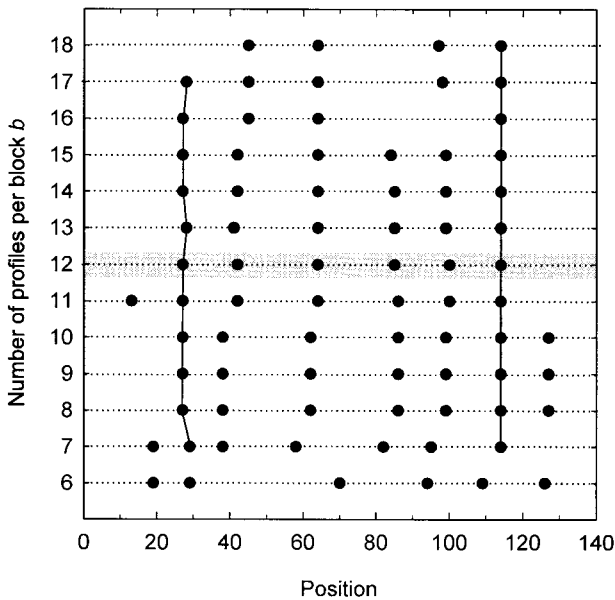


FIG. 6. Dependency of the parcellation pattern on the number of profiles per block b , demonstrated for the set of profiles shown in Fig. 4a. Abscissa: Positions j of profiles. Ordinate: b . Each dot marks a main maximum at position j in $DM^2(j)$ for a distinct value of b . Each row of dots represents a cortical parcellation pattern, $b = 12$ (shaded area) indicates the pattern in Fig. 4d. The patterns for $b = 8, 9$, and 10 are identical. If dots add up to columns, it indicates that boundaries were found at almost identical positions. The boundary between V2 and VP at positions 27/28 and the boundary between V2_{Ob γ} and V1 at position 113 are connected by vertical lines in order to demonstrate their constant positions.

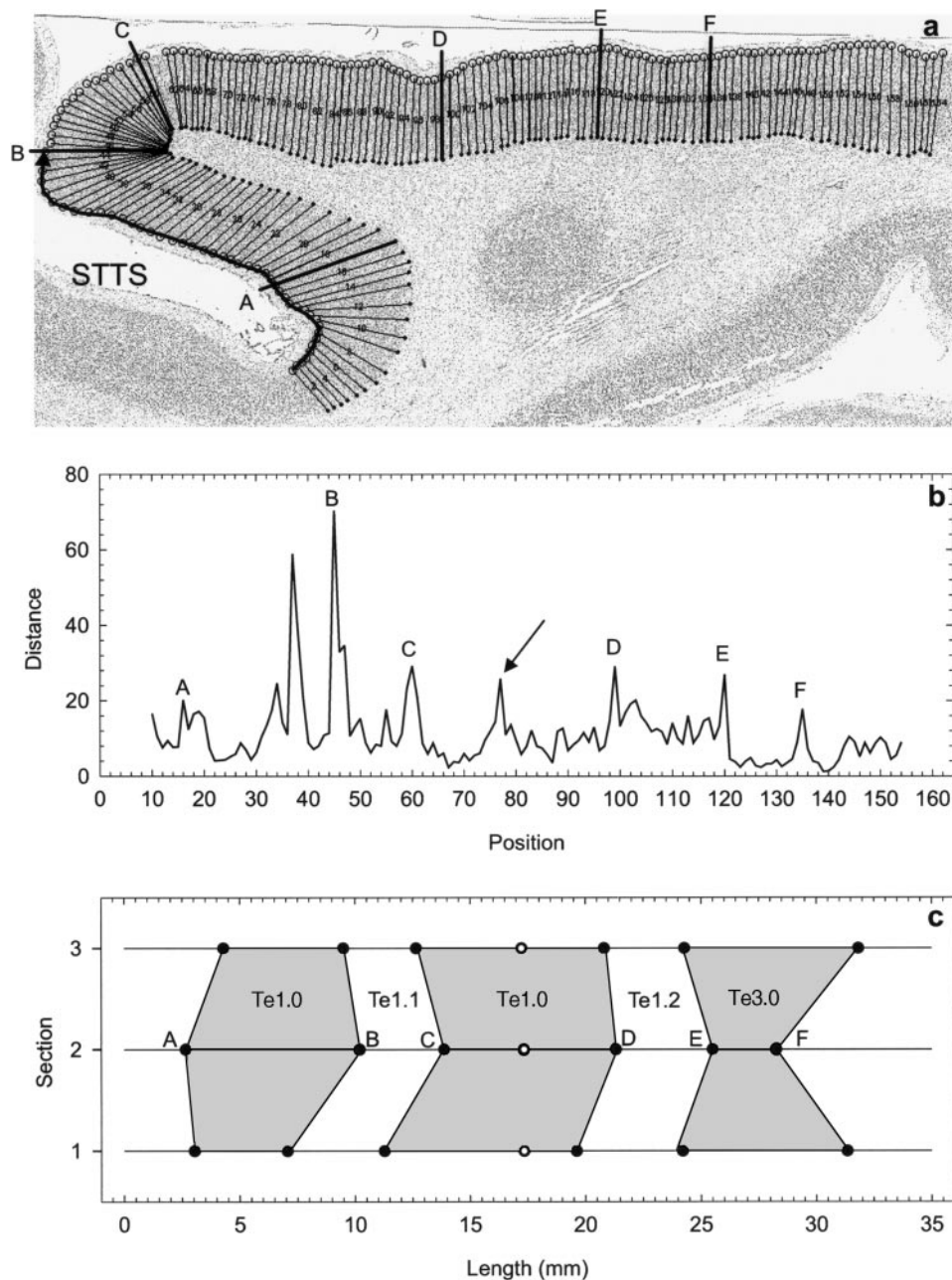


FIG. 7. (a) GLI-scan of a silver-stained, coronal section, reduced in contrast, with a set of 164 superimposed traverses which cover the human temporal areas Te1.0, Te1.1, Te1.2, and Te3.0. Starting points of traverses are marked by circles, endpoints by dots. The numbering of profile positions ($1 \leq j \leq 164$) begins at the bottom of the second transverse temporal sulcus (STTS). A to F: positions of main maxima shown in (b). (b) Mahalanobis distance function $DM^2(j)$ for $b = 10$, calculated for the set of profiles indicated in (a). A to F label main maxima which could be assigned to boundaries between cytoarchitecturally defined temporal areas. (c) Flat map representing the parcellation scheme of the section shown in (a) and its two neighbors. Ordinate: Section number. Section 2 represents the section referred to in (a) and (b), sections 1 and 3 are the two neighboring ones. Abscissa: Position of profiles. The relative positions indicated in (b) were transferred to the length values in mm beginning at the bottom of the second transverse temporal sulcus (=0 mm) and interconnecting the starting points of the traverses. An example is shown in (a) by a black line with arrowhead: It starts from position 1 and points to position 45 marked as B. The corresponding length indicated in (c) is 10.2 mm (see position marked as B in section 2). Boundaries at corresponding locations are connected by lines between 1, 2, and 3. In our sections, area Te1.0 is interrupted by Te1.1 for a short distance, B to C. This is caused by a cutting effect: Te1.1 is a small, tongue-like area extending into Te1.0 which tapers out laterally (data not shown). The boundary within Te1.0 at C to D is caused by a histological artifact affecting all three sections and is not accepted as a boundary (maximum with a small arrow in (b)).

A sampling scheme based on blocks at optimal orientation helps to avoid tangential sectioning.

Furthermore, density profiles are based on the GLI. As an estimate of volume density, it is the only stereological parameter which does not depend on the orientation of the plane of sectioning relative to the orientation of microstructural elements of the cortex (cells or cell columns). Textural parameters vary greatly if the orientation of the plane of sectioning changes and, thus, are much more sensitive to deviations from vertical sectioning than are GLI values.

Effects induced by cortical folding. Cell shape and layer width are affected by cortical folding. This was already described by von Economo and Koskinas (1925). But there is no indication that the GLI is affected as well. Changes in relative layer width affect the shape of GLI profiles. But cortical folding induces gradual changes in profile shape. By analyzing the cortical model, we have demonstrated that the main advantage of the Mahalanobis approach is its pronounced sensitivity to abrupt changes in the laminar pattern and its insensitivity to gradual changes in cytoarchitecture.

From a methodological point of view, the crown of a gyrus or the bottom of a sulcus are positions, at which the two blocks of profiles used for calculating the distance value cover two cortical sectors similar in shape and laminar pattern. If there is a perfect symmetry in relation to the axis of major orientation of the gyrus (or the sulcus) in the plane of sectioning, and no boundary is positioned in this region, both cortical sectors are identical in shape and laminar pattern. Hence, the distance value at this position is expected to be low. This corresponds to position 150 in the cortical model (Fig. 5), where the Euclidean distance functions are low, and the Mahalanobis distance functions are near zero. All applications up to now have not indicated any preferences for the detection of boundaries at the crown of a gyrus or the bottom of a sulcus. This is also illustrated by the mapping of the visual areas (Fig. 4): Profile 88 marks the bottom of the right sulcus in Fig. 4a, and the Mahalanobis distance function (Fig. 4d) at this position is low. On the other hand, if a boundary exists at or near the crown of a gyrus, it will be detected (see position 113 in Fig. 4a, boundary between V1 and OB γ).

Density measure. In the examples shown above, GLI measurements were used to quantify laminar patterns of the cerebral cortex. The GLI is a well-established stereological parameter which was introduced to neurobiological research by Haug (1956) and first measured automatically by Adhami (1973). It was also used in parcellation studies based on improved images of the laminar pattern (Sauer, 1983b; Schleicher and Zilles, 1989; Zilles *et al.*, 1980, 1986; Amunts *et al.*, 1995, 1996, 1997b; Geyer *et al.*, 1996). GLI data are not affected by local variations in staining intensity or section thickness, aspects which cannot be entirely

avoided even when adhering to highly standardized histological procedures. In addition, GLI data are closely correlated with the volume density of neurons (Wree *et al.*, 1982) and can therefore be used as a parameter to assess an important cytoarchitectonic feature of the cortex. GLI-based approaches were recently combined with the method presented here (Schleicher *et al.*, 1995; Larsson *et al.*, 1997). The existence of two hitherto unknown subareas within the human primary motor cortex was revealed, a finding which had so far escaped all visual examinations of cell-stained sections. PET findings showed that these two subareas are also functionally different entities (Geyer *et al.*, 1996).

It should be emphasized that the GLI represents only one of many parameters which describe the cytoarchitecture of the cortex. Other stereological parameters (e.g., numerical densities, cell sizes, etc.) can also be analyzed using our method. The new procedure for detecting boundaries is not even restricted to cytoarchitectonic mapping. Density profiles can be extracted and processed from any images that contain aspects of cortical layering. Density profiles have been extracted from images of myelin-stained sections in order to characterize human visual areas (Zilles and Schleicher, 1993). It is possible to combine various staining techniques and autoradiography by extracting profiles from corresponding positions in sections processed in different ways, and then integrating all features into one common vector. This has not yet been realized, but it could be implemented after adding tools required for the alignment of images from differently processed sections (e.g., Hibbard *et al.*, 1992; Toga and Banerjee, 1993).

The feature vector. The feature vector used for shape analysis of profiles is based on central moments. This approach was found to be sensitive to and specific for the detection of areal boundaries, but the features can be supplemented with or replaced by other descriptors like a Fourier series (Gillot, 1980). The selection of the 10 features outlined above is the result of an heuristic approach. We tried to reduce the number of features by analyzing their correlation matrix, but although some correlations were found to be high in several applications, there was no consistent pattern of high correlations across different cortical areas, sections, or staining techniques.

Distance measures. In a first step, the Euclidean distance was used to calculate the distance function and to define its main maxima. Some additional distance measures like the City block distance (Gower, 1985) or correlation analyses did not result in clear improvements of the algorithm when compared with the Euclidean approach. The Mahalanobis distance, however, proved to be superior, mainly due to the integration of the covariance matrix. As a result, sensitivity was improved, since not only the numerical values of the features, but also their covariance changes

at cortical boundaries. This contribution of the covariance matrix makes the amplitudes of the Mahalanobis distance difficult to interpret as a distance in feature space. In order to get an insight into the properties of the Mahalanobis distance function with regard to boundary detection, we therefore analyzed a cortical model. It clearly demonstrated an additional advantage of the Mahalanobis distance function: it is insensitive to *gradual* changes in the laminar pattern. Only relatively abrupt changes within a narrow cortical sector were detected. This means that our Mahalanobis-based approach can also be applied to cortices with gyri and sulci, where shifts in the laminar pattern are induced by cortical folding.

General aspects. The new method is a supplement to traditional procedures in the sense that parcellation schemes established using visual inspection can be validated. It still relies on careful visual screening of histological sections for sources of potential false positive boundaries which may be caused by artifacts (e.g., ruptures, folds, or local events like larger blood vessels or untypical cell clusters).

Boundaries are detected based on a multivariate shape analysis of density profiles which describe the course of only one cytoarchitectural parameter, the GLI. On the other hand, human visual perception is unrivaled in detecting even minor changes in the textural pattern of the cortex. The question arises to what extent the boundaries detected by the new procedure correspond to boundaries localized by traditional methods. We visually screened hundreds of boundaries which had been detected by the new procedure, and we were able to recognize corresponding changes in texture at these locations in all cases. This indicates that even complex, textural changes at areal boundaries are reflected by changes in the shapes of GLI profiles. Many of these minor textural changes in cytoarchitecture, however, could not be selected, evaluated, and traced across nearby sections without being predefined by the new method. The contribution of the new procedure is that it concentrates a large variety of textural descriptors into one single, but sensitive, quantitative parameter which can be processed by multivariate statistical procedures. In this sense it is not only a supplement, but an important extension of traditional procedures.

This problem raises the fundamental question of how a cortical area can be defined. Our procedure is based on the analysis of laminar patterns which are assumed to be fairly constant within a cortical area and which change at boundaries between areas. This was also the basic assumption underlying the classical cytoarchitectonic studies by Brodmann (1909) and others (references in Zilles *et al.*, 1978). Several studies have shown the existence of microstructurally defined cortical areas which are not readily detectable in the microscope (e.g., Zilles *et al.*, 1982). Microstructural/functional relation-

ships are the subject of intensive investigations, and for some areas in the human cortex this relationship has already been established (Geyer *et al.*, 1996; Roland and Zilles, 1994).

The algorithm for cortical parcellation was designed to detect microstructural boundaries by detecting changes in the laminar pattern of neocortical images. The classification of these boundaries is a second and subsequent step that needs additional information, collected, e.g., by comparing it with classical architectonic descriptions or looking for it in other stains or by comparing the results with parcellations based on functional data. In other words, the new algorithm detects microstructural boundaries, but does not provide classification or functional interpretation.

Up to now, differences in microstructure representing boundaries between areas could not be systematically quantified and processed. Various qualitative criteria were established which were not consistent among different investigators, even with respect to the same boundaries. The algorithm described here has clearly shown that it can locate microstructural boundaries. Since these boundaries are defined using an observer-independent approach, they are reproducible and can be described by numerical data. The statistical nature of the procedure and the finding that the boundaries can be traced in a series of sections is clear evidence that they are not random events. This is a major improvement compared to the classical approach which is based on pure visual inspection of histological specimens. It offers a new method for the anatomical parcellation of the cerebral cortex, an important prerequisite if structural data is to be correlated with data acquired by functional imaging techniques.

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